"I've Seen How This Goes": Characterizing the Diversity of LLM Generations and Human Writing via Progressive Conditional Surprise

In-context learning of pretrained models can be used to directly measure the diversity of LLM generations and human writing.

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Why measure diversity?

The possibility for diverse outputs is necessary, but not sufficient, for creativity. Researchers routinely need to compare the diversity of outputs across:

- elicitation methods during propensity and capability evaluations, to get a complete picture of what the model is capable of and what it might choose to do
- post-training stages (mode collapse)
- decoding strategies (what does temperature trade off against?)
- human–AI hybrid pipelines (how can AI be used to amplify rather than weaken human creativity?)

Existing metrics rely on embedding distances or surface-level n-gram statistics, both of which lack a human-like ability to recognize arbitrary patterns.

The idea: in-context learning as a lens

We exploit the in-context learning capability described by Brown et al. (2020), applied here as a measurement lens rather than as a few-shot-prompting setup. As θ sees more responses, its in-context learning picks up on shared modes, stylistic regularities, and topic patterns.

If responses from a policy π are diverse, seeing one should not help θ predict the next, whereas if they are repetitive or constrained to a few modes, conditioning on earlier responses should sharply reduce θ's surprise at later ones.

Diversity is therefore a property of (responses, prompt, scoring model). The metric measures diversity as θ perceives it.

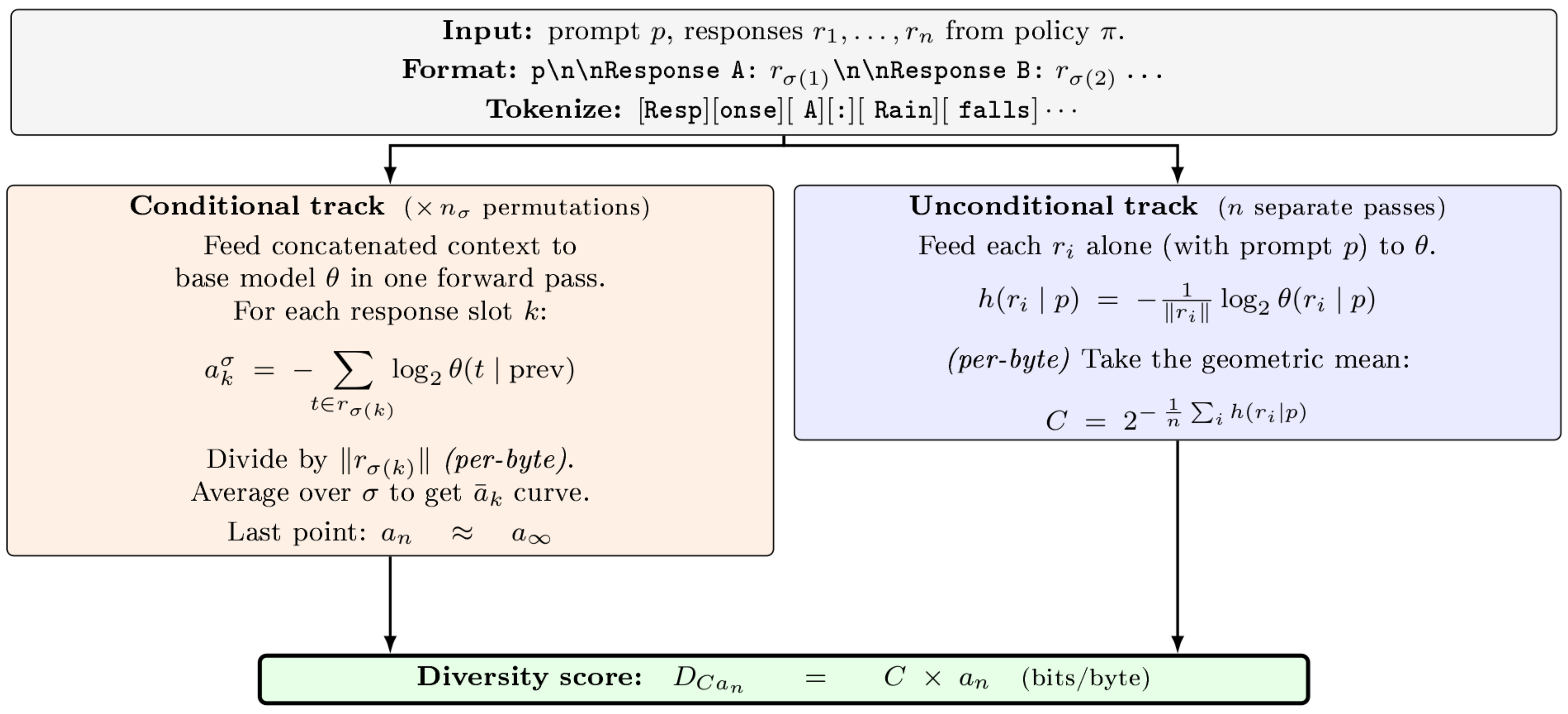
Takeaways

- A base model's in-context learning detects similarities between arbitrary numbers of responses in a **single forward pass per permutation**.
- Measuring diversity with ICL requires no special-purpose training, datasets, or predetermined aspects of diversity.
- The same pipeline scores AI samples and human-written sets, and tracks post-training diversity loss on OLMo-2-7B.

Read the full paper

Code & data

Method: the Decan score $D_{Ca_n} = C \times a_n$



Decan is read off the per-token log-probabilities of a base model θ, in one forward pass per permutation.

Progressive conditional surprise $a_k = -\frac{1}{\|r_k\|} \log_2 \theta(r_k \mid r_{<k}, p)$

- How surprised θ is by response k given the previous $k-1$. It is averaged over random response orderings (permutations). The curve of a_k values characterizes diversity, and scalar summaries attempt to capture its shape.

Residual surprise $a_n \approx a_\infty = \frac{1}{\|r_n\|} \left(H_\theta(r_n \mid p) - I_\theta(r_n; r_{<n} \mid p)\right)$

- The curve's last point: the response's own surprise minus what the others reveal about it. High a_n indicates that responses remain surprising even after seeing the rest.

Coherence $C = 1/\mathrm{PPL}_\theta(\pi, p)$

- Reciprocal perplexity suppresses the diversity score of incoherent sets because noise should not make the metric consider the set to be more diverse.

Decan $D_{Ca_n} = C \times a_n$

- Bits of surprise per byte left after θ learns what it can from the other responses, weighted by coherence.

Human writing diversity evaluation

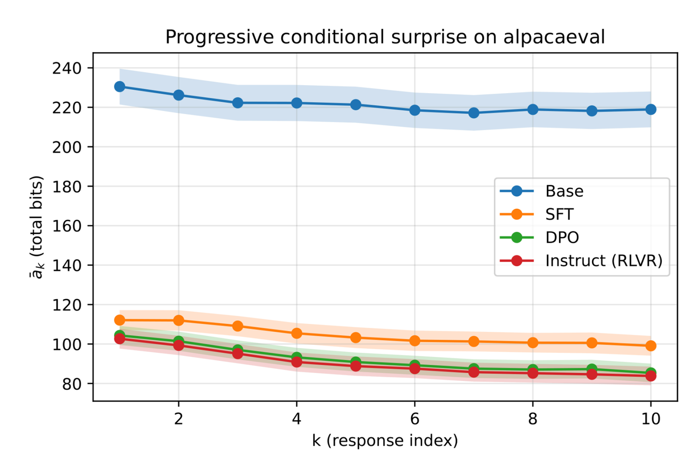
We evaluate on Tevet and Berant's human diversity benchmark. The response sets are written by Mechanical Turk workers, with the high/low diversity label fixed by each dataset's construction protocol.

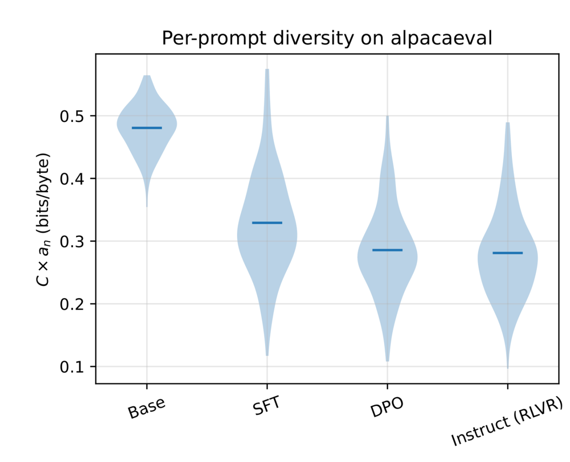
Across the binary tasks Decan clearly tracks human diversity judgments, reaching close to the sentence-embedding baseline of SentBERT, but still 3.7–21.0% behind in OCA. Future work can attempt to close the gap by using other scalar summaries of the a_k curve or fine-tuning the base model.

	prompt_gen		resp_gen		story_gen	
Metric	ρ	OCA	ρ	OCA	ρ	OCA
ConTest (200, with_hds)						
$C \times a_n$ (ours)	+0.584	0.785	+0.391	0.668	+0.686	0.828
a_n (ours)	+0.444	0.715	+0.274	0.641	+0.387	0.684
C (ours)	+0.214	0.625	−0.001	0.555	+0.247	0.632
SentBERT	**+0.682**	0.815	**+0.591**	**0.791**	**+0.770**	**0.896**
BERTsts	+0.646	**0.820**	+0.463	0.714	+0.601	0.780
distinct-n	+0.333	0.675	+0.346	0.677	+0.573	0.772
McDiv_nuggets (∼1K, no_hds)						
$C \times a_n$ (ours)	+0.636	0.785	+0.345	0.649	+0.317	0.634
a_n (ours)	+0.487	0.705	+0.225	0.619	+0.124	0.557
C (ours)	+0.138	0.567	+0.082	0.545	+0.251	0.643
SentBERT	**+0.728**	**0.850**	**+0.532**	**0.758**	**+0.633**	**0.803**
BERTsts	+0.683	0.830	+0.393	0.676	+0.344	0.638
distinct-n	−0.003	0.514	−0.002	0.507	−0.002	0.510
McDiv (full, no_hds, ∼2K)						
$C \times a_n$ (ours)	+0.729	0.846	+0.500	0.724	+0.523	0.717
a_n (ours)	+0.617	0.781	+0.432	0.698	+0.402	0.668
C (ours)	+0.138	0.565	−0.007	0.512	+0.171	0.594
SentBERT	**+0.796**	**0.897**	**+0.678**	**0.830**	**+0.753**	**0.867**
BERTsts	+0.780	0.893	+0.614	0.781	+0.571	0.740
distinct-n	+0.476	0.746	+0.517	0.738	+0.535	0.744

Spearman ρ and OCA against human diversity labels. $C \times a_n$ and a_n are ours (Qwen2.5-3B, 50 permutations, per-byte); baselines are from Tevet and Berant. Best per task in bold.

AI writing evaluation: Post-training mode collapse



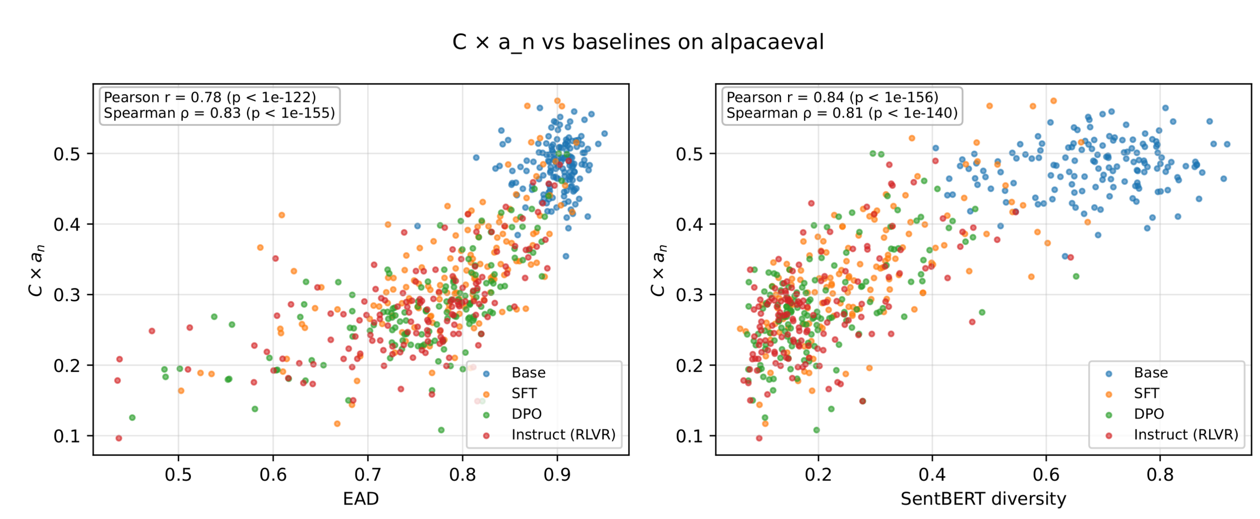


On the OLMo-2-7B post-training pipeline, Decan decreases monotonically across the base, SFT, DPO, and RLVR stages, detecting the type of diversity loss that creative-writing applications care about.

base 0.481 → SFT 0.329 → DPO 0.286 ≈ RLVR 0.281

All three pre-registered contrasts are significant after Bonferroni correction.

AI writing evaluation: Agreement with existing metrics



Per-prompt Decan also agrees with the lexical (EAD) and semantic (SentBERT) baselines on the same length-matched subset, confirming the metrics see the same diversity-loss signal.

References. Tevet & Berant. Evaluating the Evaluation of Diversity in NLG. EACL 2021. · Brown et al. Language Models are Few-Shot Learners. 2020. · Team OLMo et al. 2 OLMo 2 Furious. 2024. · Reimers & Gurevych. Sentence-BERT. 2019.

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